

Protein Structure Publication Figures · Skill Usage Guide (English)

A **protein-figure** skill built on PyMOL-PUB (Bioinformatics 2024, btae139) and the PyMOL 2.5.0 open-source edition.
Ships with 10 direct-cmd style recipes + 3 standard-class templates, all validated against 1AY7.

1. What This Skill Does

Renders protein/nucleic-acid 3D structures (PDB/PDBx) into journal-ready figures: white background, 300–600 DPI, in-figure labels default to English, publication-grade palettes. It covers three needs:

- Journal-spec layouts: Nature/Science/Cell/PNAS/ACS/Oxford/PLOS/IEEE fonts, column widths, minimum DPI handled automatically.
- Highlighting regions: binding sites, mutations, key residues, active centers with custom palettes.
- Comparison & mechanism figures: dual-structure RMSD overlays, residue-pair distances / polar contacts, protein–protein interfaces, NMR/MD ensembles.

2. Environment & Installation (one-time)

Hard constraint: Python 3.9–3.11 + PyMOL must equal 2.5.0 (open source) + **PyMOL - PUB**.

Recommended conda-forge route (validated; env name **pymol_pub** is just a convention):

```
conda create -n pymol_pub python=3.11 -c conda-forge -y conda install -n pymol_pub -c conda-forge "pymol-open-source=2.5.0" -y conda run -n pymol_pub python -m pip install PyMOL-PUB
```

Verify:

```
conda run -n pymol_pub python -c "import pymol2; print(pymol2.__version__)" # expected: 2.5.0
```

Note: conda-forge's **pymol-open-source** is the reliable source for 2.5.0; the
gohkle third-party wheels and PyPI's **pymol-open-source** (only 3.x alphas) cannot
deliver 2.5.0 — avoid both.

3. Two Rendering Paradigms

Paradigm	What	When
A: standard classes (molpub)	HighlightStructureImage / PropertyStructureImage / Figure	Journal layouts, multi-panel publication figures, RMSD property shading
B: direct cmd (entry_class: null)	from pymol2 import PyMOL + hand-written cmd sequence	Custom palettes / soft-light / flat look that exceed the class API (all 10 direct-cmd recipes)

Calling-layer rule: **mol = PyMOL(); mol.start(); cmd = mol.cmd**, end with **mol.stop()**. Do not use **import pymol** as the calling layer.

2.5.0 Strict-API Cheatsheet (top 6 gotchas)

- Custom palettes via **cmd.set_color(name, [r,g,b])**, not **set("color", ...)**.
- White background via **cmd.bg_color("white")**, not **set("background_color", ...)**.
- Cleanup via **cmd.hide("everything", "hetatm") + cmd.hide("everything")**; first arg of **hide** must be a valid representation (**everything** ok, **all** not).

- Merge multi-residue selections with `cmd.select` + `cmd.extend` step-by-step; named selections reject `+/or` concatenation.
- `cmd.spectrum("bfactor", ...)` is not recognized → use `resi` (or `b` if the PDB has a B column, `plddt` if it has a pLDDT column).
- Keep object names off reserved words (`a/b` get an underscore appended); prefer `s1/s2`.

4. Style Recipe Overview (10 direct-cmd + 3 standard)

4.1 Direct-cmd style recipes

Recipe	Look	Typical use
<code>pastel_cartoon_surface</code>	Per-chain cartoon + milky translucent shell	Multi-chain pastel overview
<code>ss_milky_surface</code>	Per-secondary-structure cartoon + neutral gray shell	Emphasize α/β /loop
<code>ss_frosted_surface</code>	Per-secondary-structure cartoon + same-color frosted shell	Richer secondary-structure look
<code>ss_only_surface</code>	Pure secondary-structure cartoon, no shell	Clean, unobstructed lines
<code>residues_only</code>	Hide whole protein, show only key-residue sticks	Focused local site
<code>spectrum_resi</code>	Cartoon graded blue-white-red by residue index + CA spheres	B-factor / pLDDT / sequence-property spread
<code>active_site_highlight</code>	Key-residue sticks + translucent cartoon environment + polar-contact dashes	Active-site mechanism view
<code>distance_contact</code>	Two-residue sticks + CA spheres + distance dashes (<code>dist mode=2</code>)	Binding-site distance annotation
<code>goodsell_style</code>	Sphere model, per-chain Goodsell pastel, flat no-shadow white render	Journal-cover-grade illustration
<code>mutation_site</code>	Single mutation-residue sticks + CA spheres + env cartoon + salmon hue	Mutation-site analysis
<code>surface_render</code>	Per-chain independent surface/mesh/dots + transparency	Molecular-surface shape / pocket
<code>ensemble_overlay</code>	Multi-model state rainbow gradient + transparency	NMR/MD flexibility
<code>dual_align_rmsd</code>	Two structures CA-aligned + RMSD + overlaid cartoon	WT vs mutant / expected vs predicted
<code>ppi_interface</code>	Two chains <code>within</code> radius → interface residues + translucent interface surface	Protein–protein interface

Note: `_registry.md` lists 17 template rows (3 standard classes + 10 direct-cmd + scene-level entries); the table above lists the 14 independently renderable recipes.

4.2 Standard-class templates (molpub)

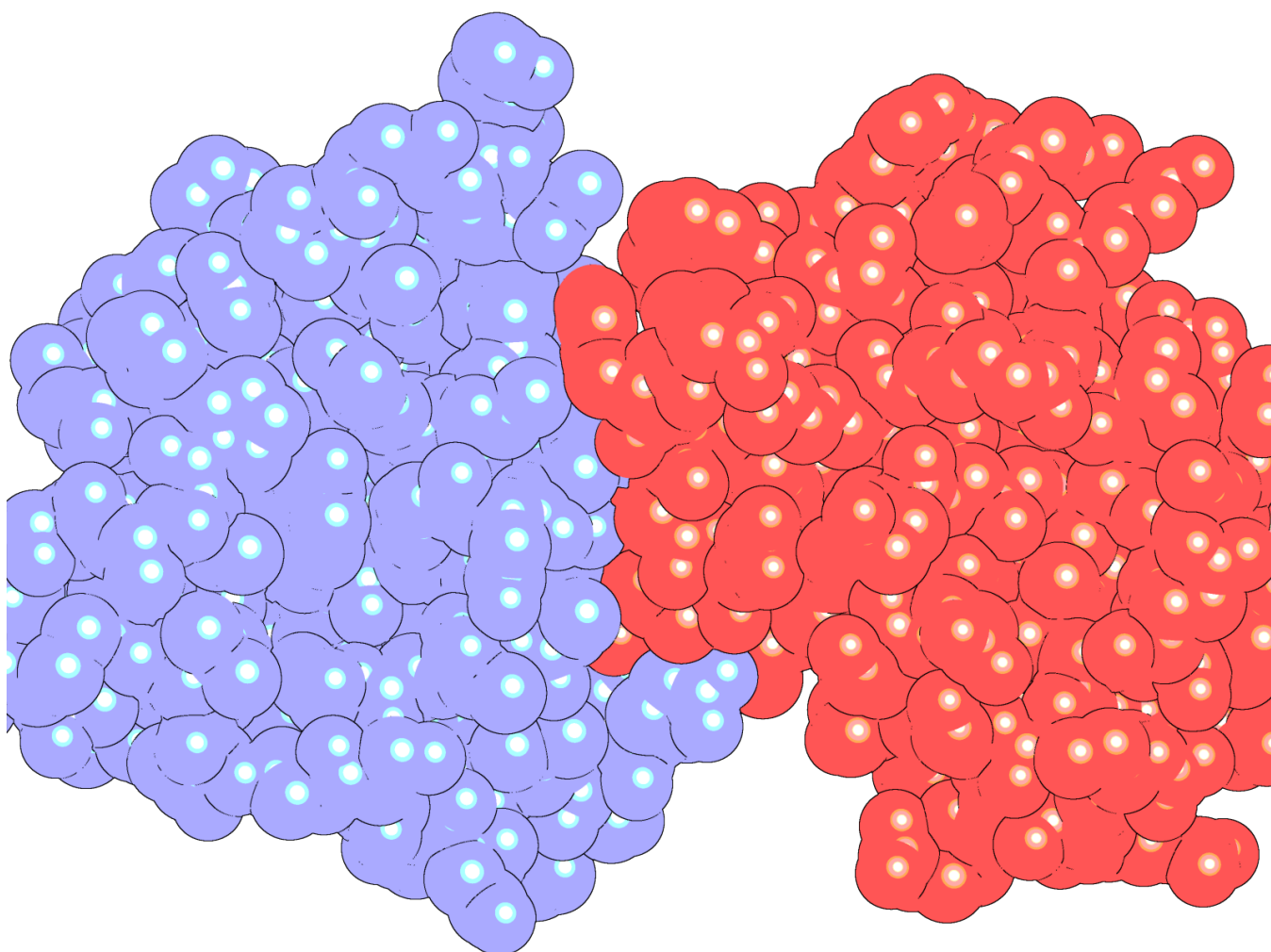
Template	Entry class	Purpose
nature_highlight	HighlightStructureImage	Multi-chain highlight, emphasize binding site / mutation / segment
rmsd_compare	PropertyStructureImage	Dual-structure RMSD/property overlay + putty cartoon residue difference
science_publication_layout	Figure	Multi-image + panels + text + widgets into a publication Figure

5. Case 1: Goodsell cover-grade illustration (direct cmd)

Goal: render 1AY7's two chains as the "flat pastel spheres" look common on journal covers.

```
# goodsell_style.py (validated on 1AY7) from pymol2 import PyMOL PDB, OBJECT = "1AY7.pdb", "m"
CHAIN_COLORS = {"A": "gs_blue", "B": "gs_red"} # only chains that exist in the PDB mol = PyMOL();
mol.start(); cmd = mol.cmd cmd.load(PDB, OBJECT, quiet=1) if int(cmd.count_atoms("all")) == 0: raise
SystemExit("LOAD FAILED: 0 atoms") # Flat: high ambient, zero reflect, no shadow for k, v in
{"ray_trace_mode":3, "ray_shadows":0, "depth_cue":0}.items(): cmd.set(k, v) cmd.set("ambient", 1.0);
cmd.set("direct", 0.0); cmd.set("reflect", 0.0) cmd.bg_color("white"); cmd.orthoscopic(1) # Custom
colors must be defined in the same instance for name, rgb in {"gs_blue":[0.565,0.714,0.812],
"gs_red":[0.855,0.475,0.427]}.items(): cmd.set_color(name, rgb) cmd.remove("m and resn HOH");
cmd.hide("everything", "hetatm"); cmd.hide("everything") cmd.show("spheres", OBJECT) for c, col in
CHAIN_COLORS.items(): cmd.color(col, "m and chain %s" % c) cmd.orient(OBJECT); cmd.ray()
cmd.png("1AY7_goodsell.png", width=1600, height=1600, dpi=300, quiet=1) mol.stop()
```

Result (1AY7, 1600×1600, 300 DPI):



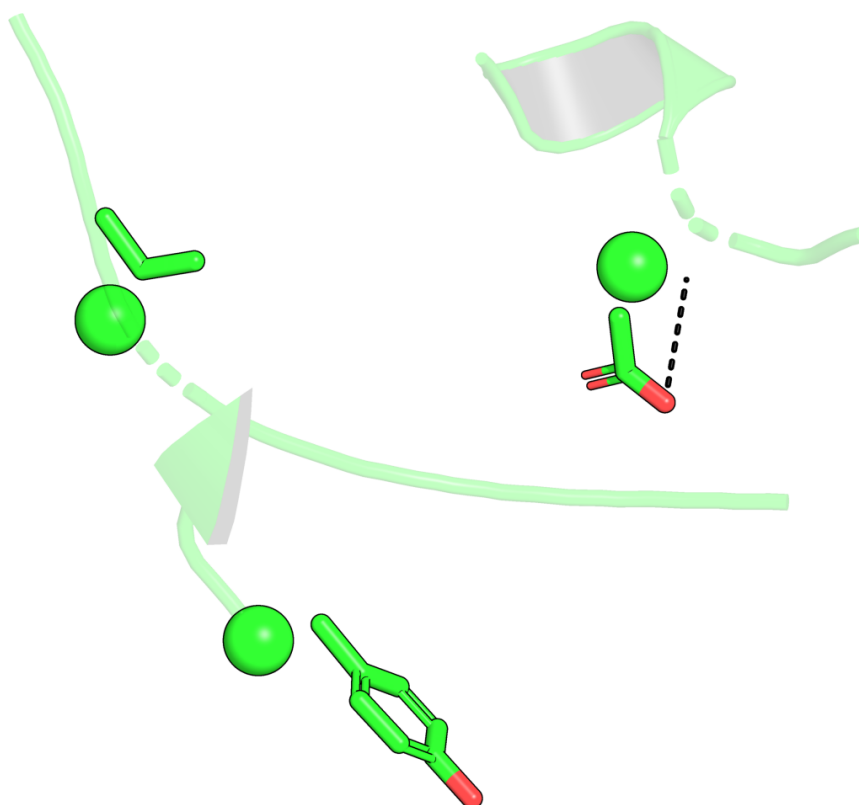
Goodsell flat pastel spheres (per-chain, white bg, no shadow)

6. Case 2: Active-site close-up (direct cmd)

Goal: blow up a handful of key residues as sticks, keep a translucent cartoon environment, and draw polar-contact dashes between residues.

```
# active_site_highlight.py key fragment KEY = ["A/35", "A/52", "B/44"] # key residues (with chain
prefix) cmd.show("cartoon", "m"); cmd.color("gray80", "m"); cmd.set("cartoon_transparency", 0.7)
merged = "msite" for i, r in enumerate(KEY): cmd.select("tmp%d" % i, r) cmd.extend(merged, "tmp%d" % i)
# named selections merged via select+extend (no + concat) cmd.show("sticks", merged);
cmd.show("spheres", "byres merged and type c") cmd.color("gs_red", "byres merged and type c") for i in
range(len(KEY)-1): # polar-contact dashes cmd.dist("d%d" % i, "%s" % KEY[i], "%s" % KEY[i+1], mode=2,
color="yellow") cmd.orient(merged); cmd.zoom("m", 1.4); cmd.ray() cmd.png("1AY7_active_site.png",
width=1600, height=1600, dpi=300, quiet=1)
```

Result:



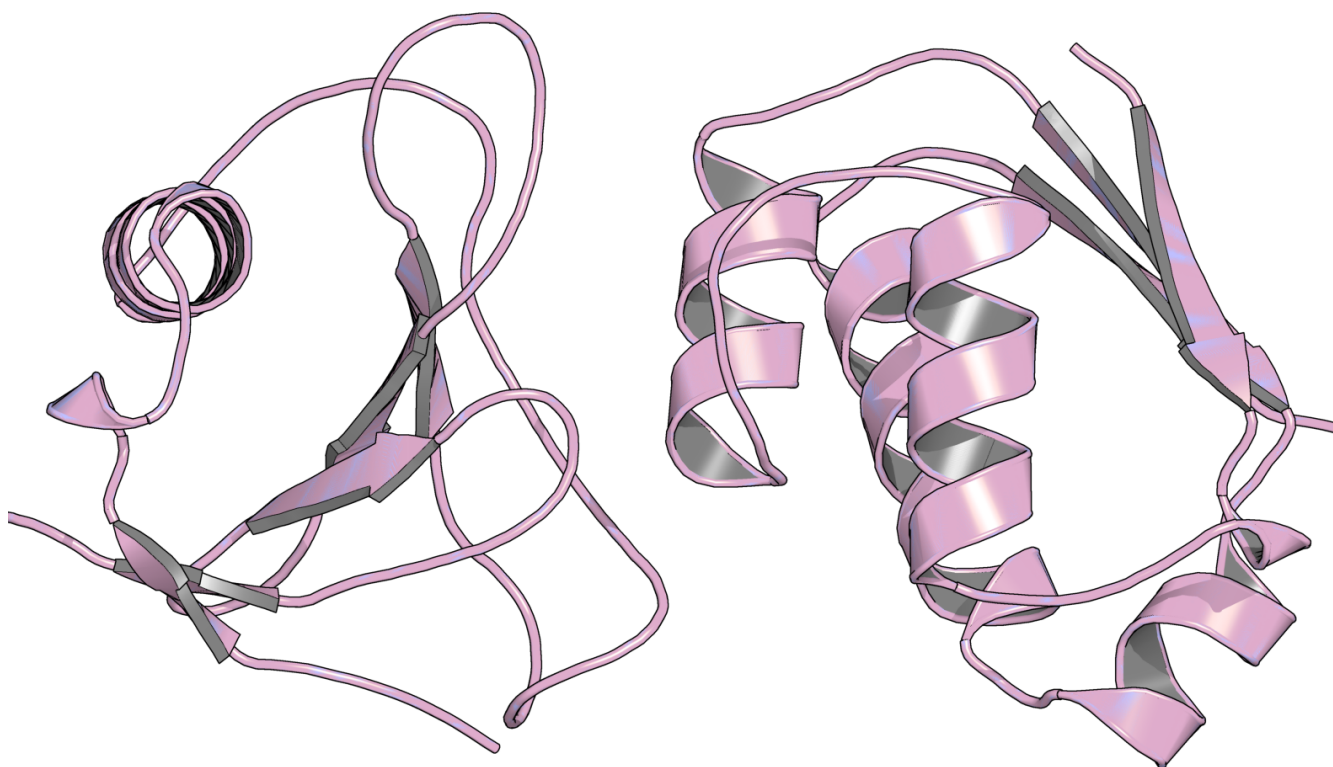
Active-site close-up (sticks + translucent cartoon env + polar contacts)

7. Case 3: Dual-structure RMSD overlay (standard class)

Goal: expected vs predicted overlay with a putty cartoon rainbow map of residue differences.

```
from molpub import PropertyStructureImage img =
PropertyStructureImage(structure_paths=["expected.pdb", "predicted.pdb"])
img.set_shape(representation_plan=[("model:predicted", "cartoon"), ("model:expected", "cartoon")])
img.set_state(rotate=[0,60,255], inner_align=True, target="expected")
img.set_color(target="model:predicted", color_map="rainbow", edge_color="0x000000",
gauge_strengthen=True) # cartoon only img.save(save_path="1AY7_align.png", width=1800, ratio=0.5)
img.close() # always close to free the PyMOL process when batching
```

Result (self-overlay demo):



Dual-structure overlay + RMSD

8. Case 4: Journal layout (standard class Figure)

Goal: compose several structure images + rotation widget icons + text into a Nature two-column publication figure.

```
from molpub import Figure, obtain_widget_icon
obtain_widget_icon(save_path="arrow.png",
widget_type="arrow", params={"degree": 90, "color": "black", "width": 0.02}) fig =
Figure(manuscript_format="Nature", occupied_columns=2, aspect_ratio=(606,358), mathtext=False,
row_number=2, column_number=2) fig.set_image(image_path="1AY7_goodsell.png", layout=(1,1,1))
fig.set_image(image_path="arrow.png", locations=[0.85,0.8,0.1,0.1], transparent=True)
fig.set_text(annotation="(a)", locations=[0.02,0.97,0.08,0.03]) fig.save_figure("final_figure.png") #
sub-figure dpi must be >= journal minimum_dpi
```

9. Publication Quality Constraints (check these)

Journal	Font	Min DPI	Max cols
Nature	Arial	300	2
Science	Helvetica	300	3
Cell	Arial	300	2(3)
PNAS	Helvetica	600	2
ACS	Arial	600	2
Oxford	Arial	350	2

- `save` defaults to `dpi=1200`; `Figure.minimum_dpi` is journal-driven — if a sub-figure PNG's real DPI is below it, `paste_bitmap` raises `ValueError`.
- Cell must be passed `column_format=2` or `3`.
- `set_image` accepts only `.png`; rasterize SVG/PDF widget icons before embedding.

10. Advanced Rendering Parameters (OPIG)

Parameter	Effect	Suggested
<code>surface_quality</code>	Surface normal / mesh precision	10–40 (not settable in 2.5.0 strict mode — known gotcha)
<code>cartoon_sampling</code>	Cartoon sampling density	2–4
<code>cartoon_line_width</code>	Cartoon outline width	0.6–2.0
<code>ray_quality</code> / <code>ray_simplify</code>	Ray-trace quality / simplification	0–1
<code>ray_opaque_background</code>	Opaque white background	1 (stable when headless, no GPU)
<code>field_of_view</code>	Perspective distortion	default 30–45 for publication

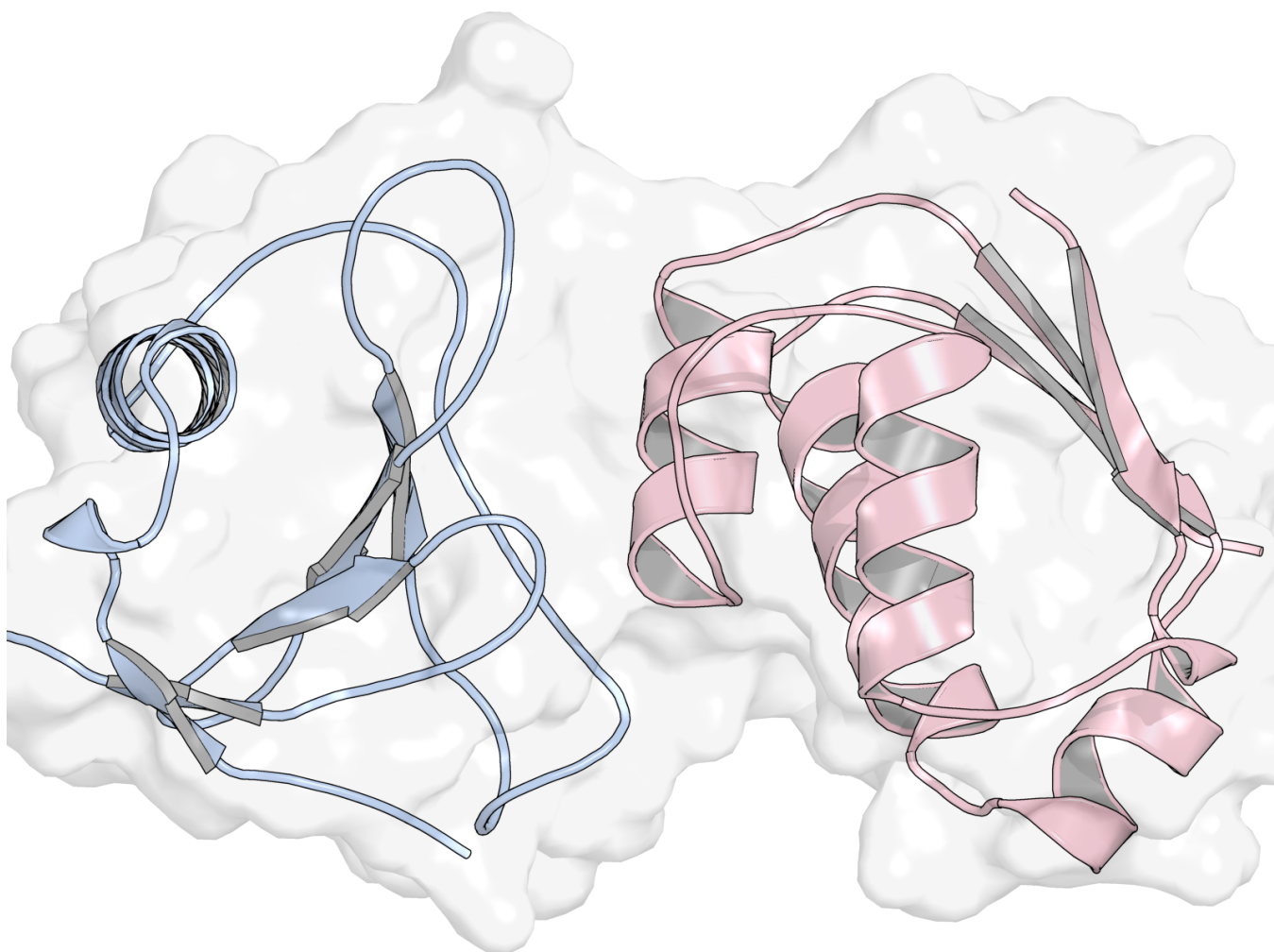
One-line publication cartoon: `cmd.preset("publication", "m")` (works in 2.5.0; `preset.pretty` is degraded in 2.x).

11. Deliverables & Preferences

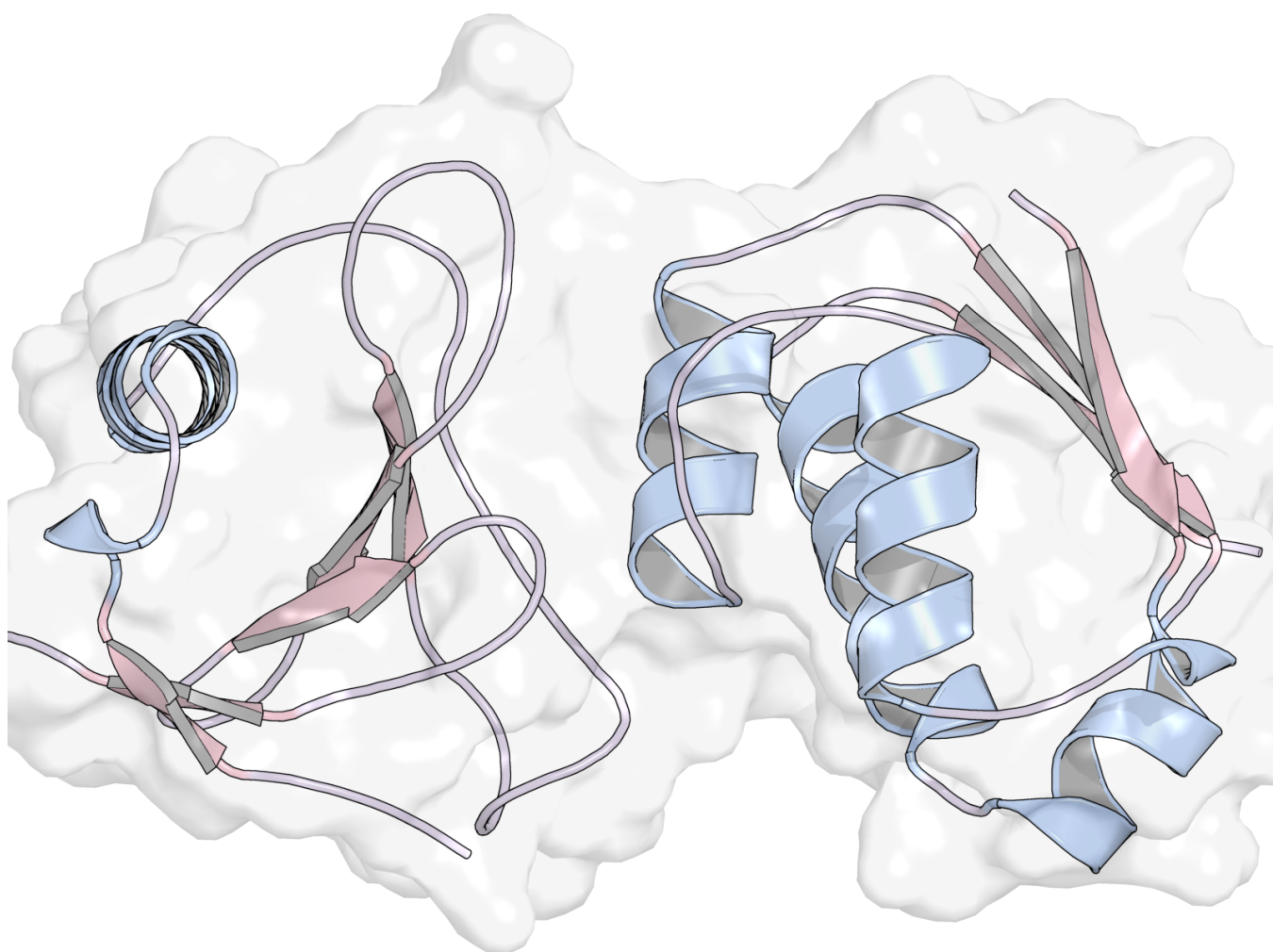
- In-figure labels default to English (panel A/B/C, axes, arrow notes, colorbars; gene/species names stay English); document body & captions in Chinese.
- Deliverables: PNG (300–600 DPI) + optional PDF/SVG; for journals requiring TIFF, convert from PNG with PIL.
- Rhythm rule: no recipe enters `_registry.md` before a verifiable sample render — first render a real PDB, PIL-validate (exists / size / non-blank), then register.

12. Sample Gallery (all 1AY7, 1600×1600, 300 DPI)

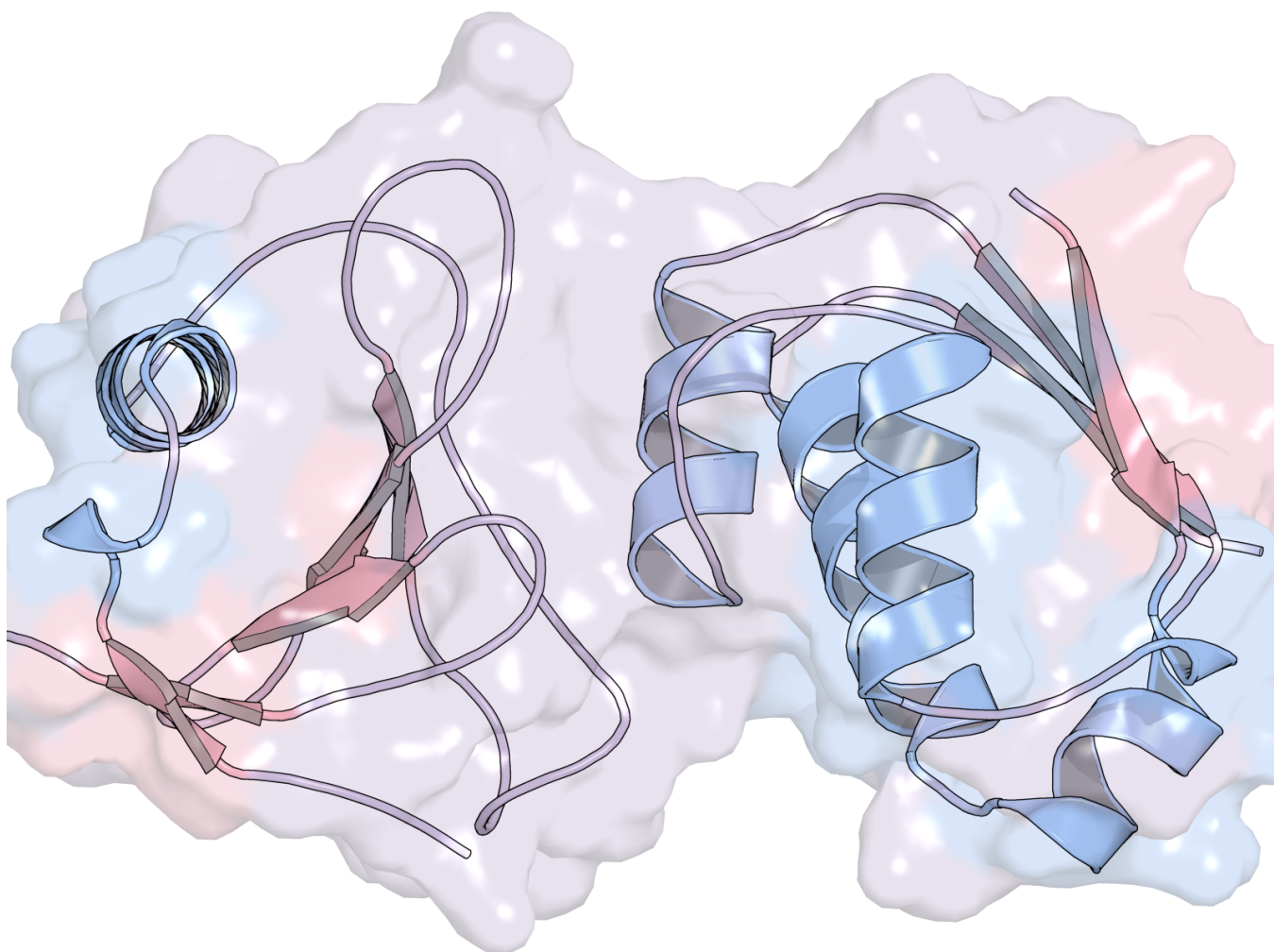
Direct-cmd recipe results



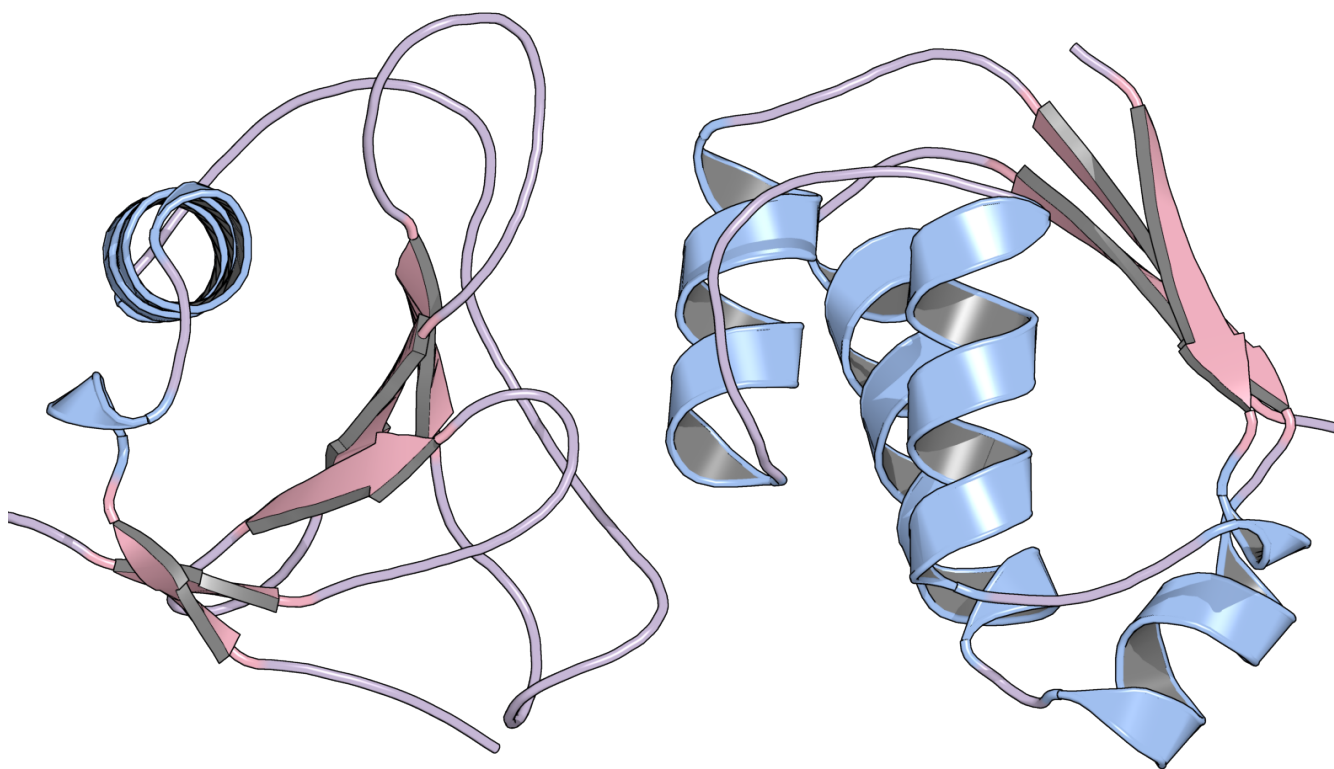
pastel per-chain cartoon + milky shell



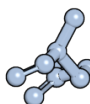
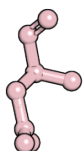
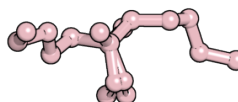
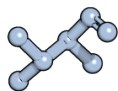
ss_milky secondary-structure cartoon + neutral shell



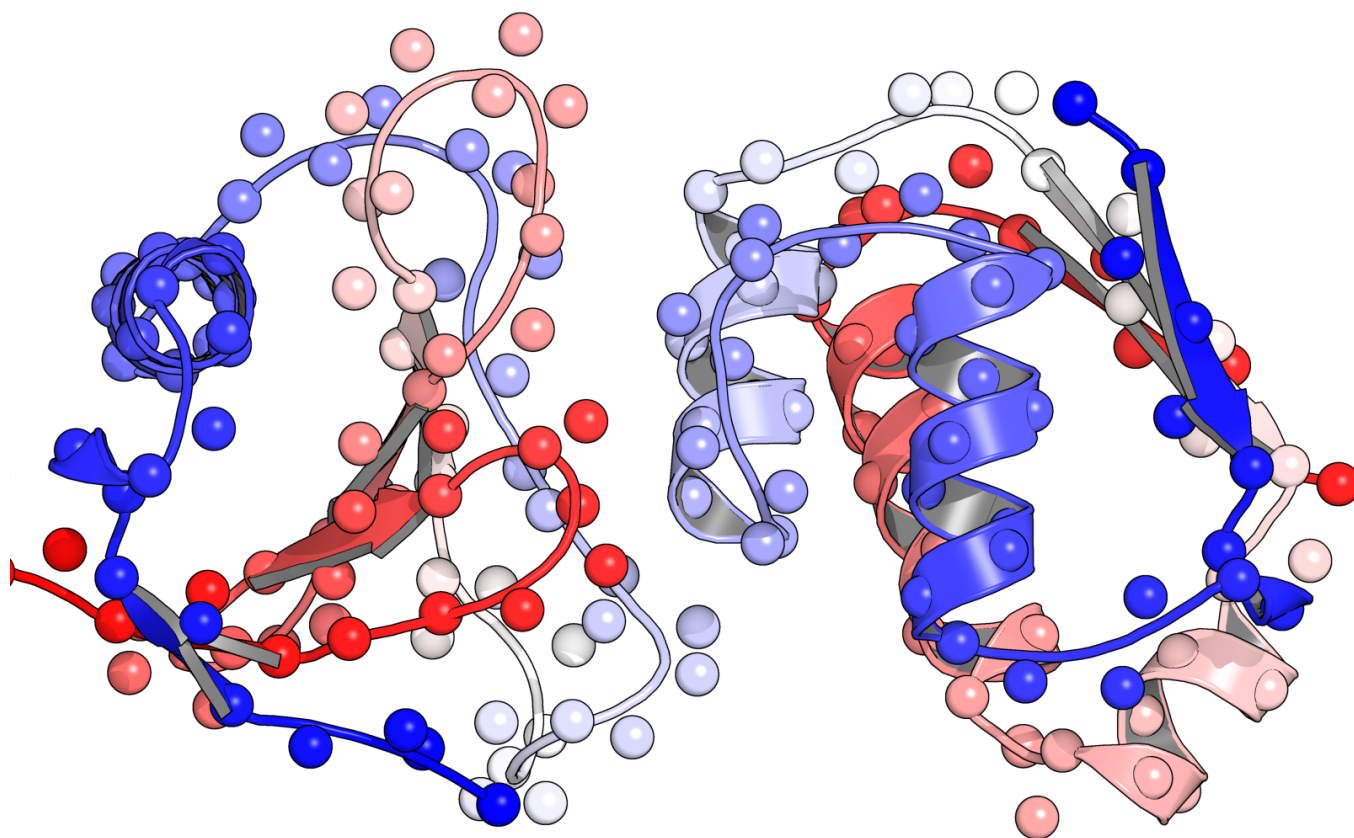
ss_frosted secondary-structure cartoon + same-color frosted shell



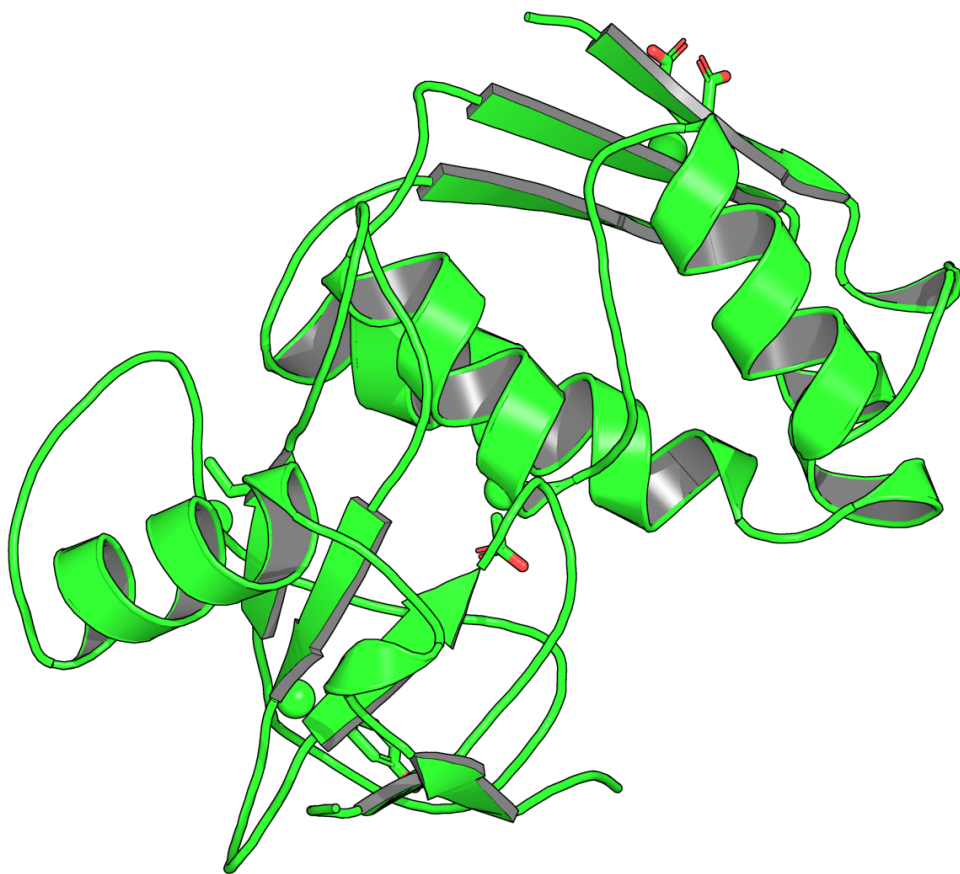
ss_only pure secondary-structure cartoon



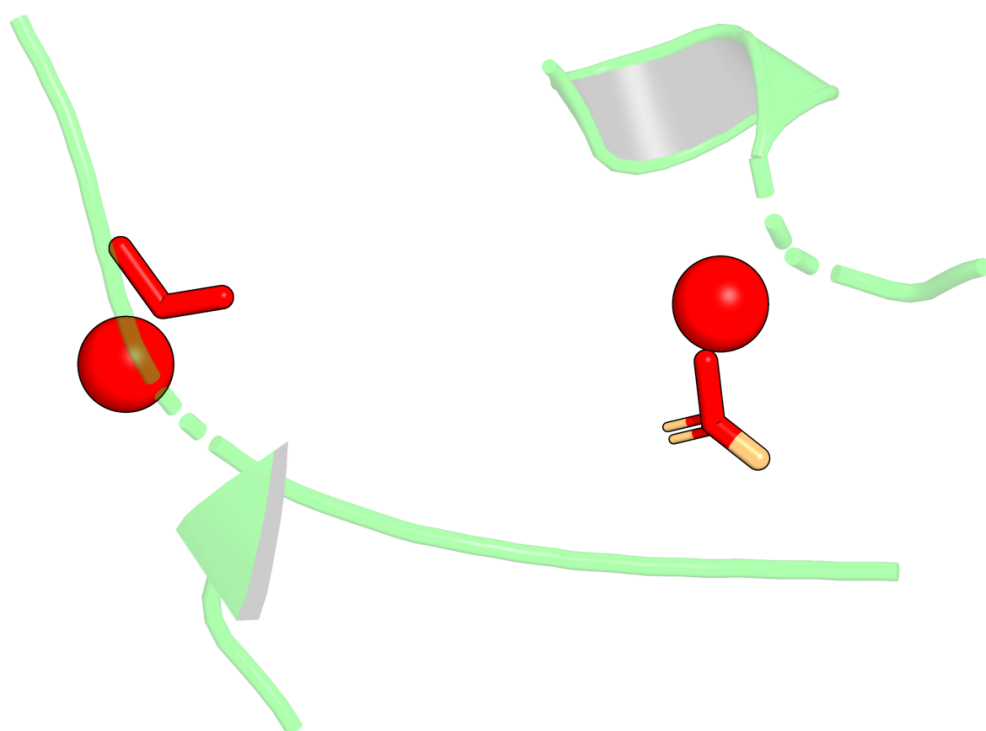
residues_only key-residue sticks close-up



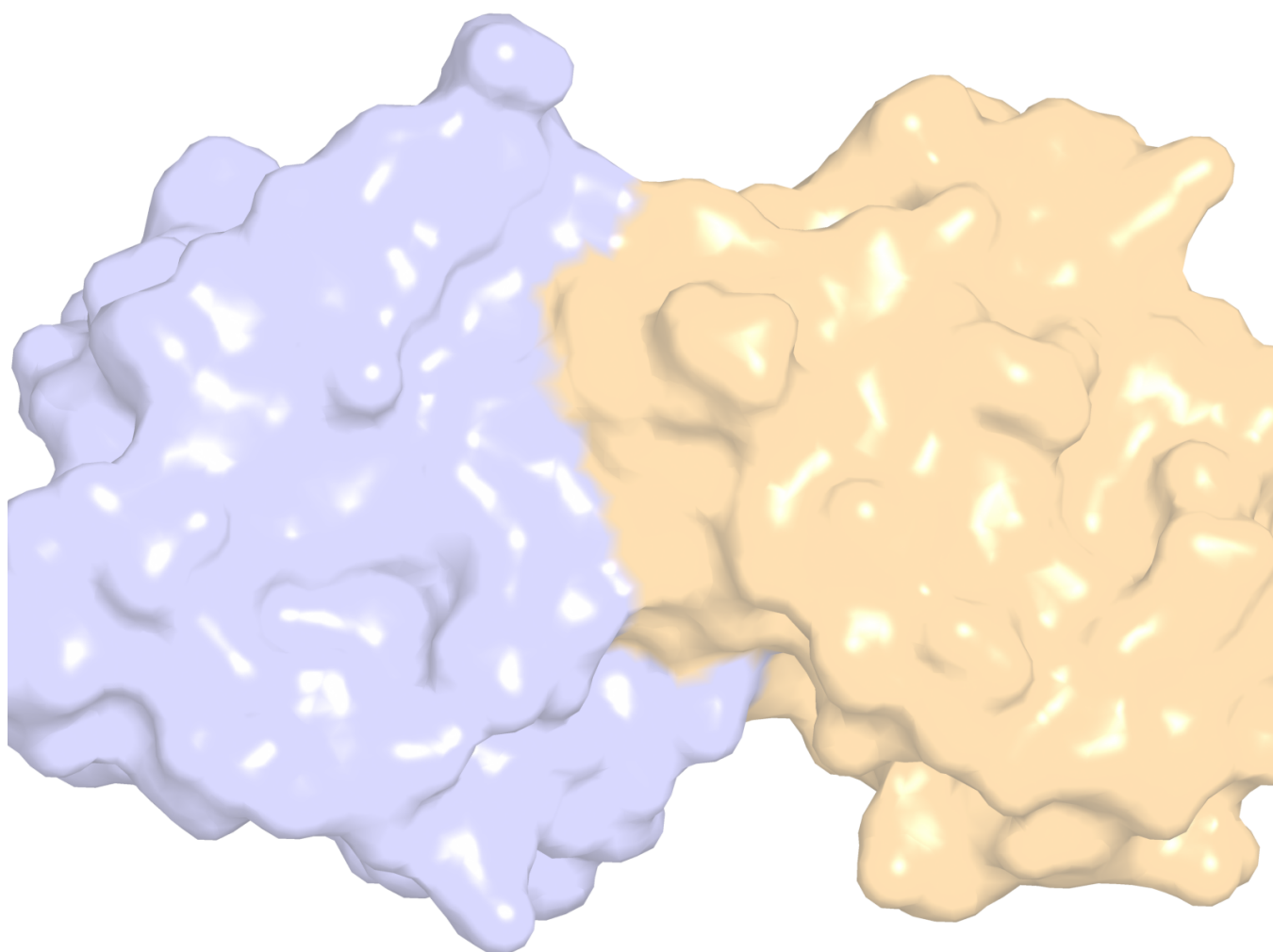
spectrum residue-position blue-white-red gradient



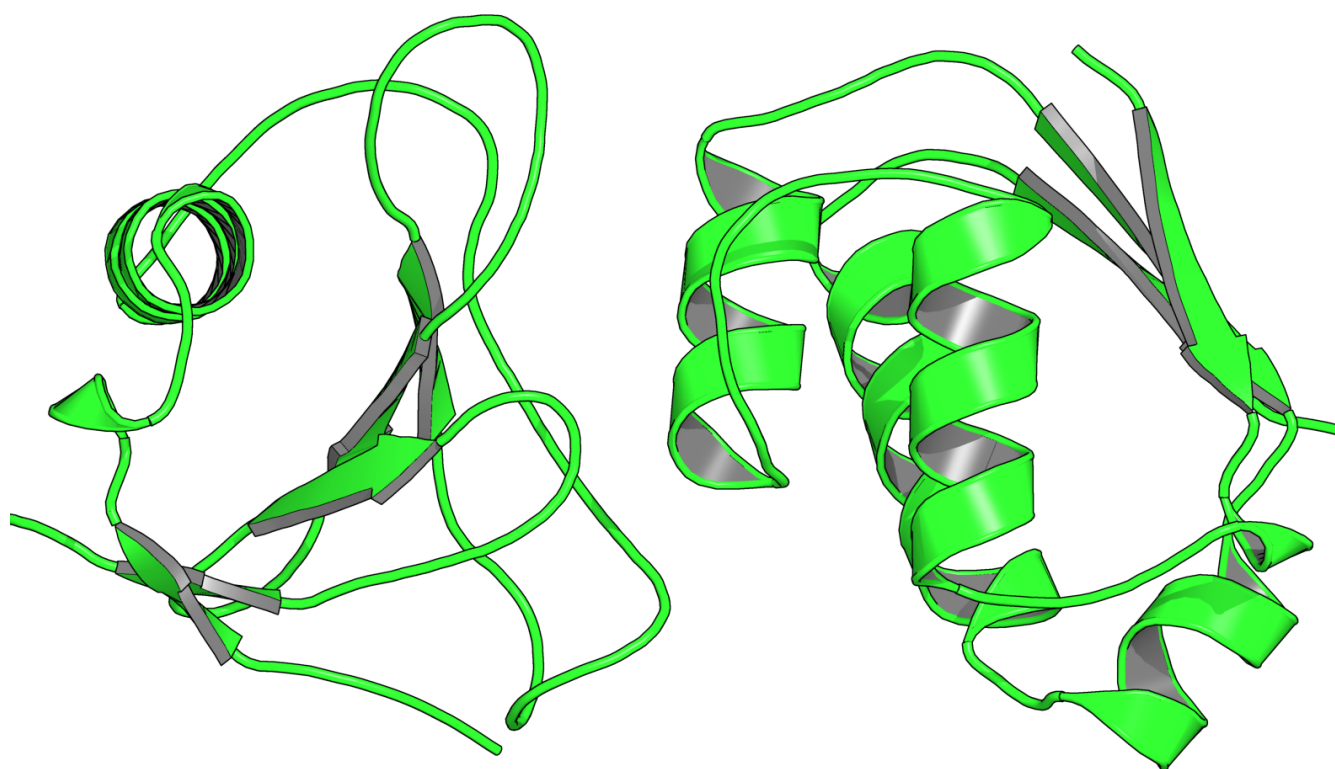
distance residue-pair annotation



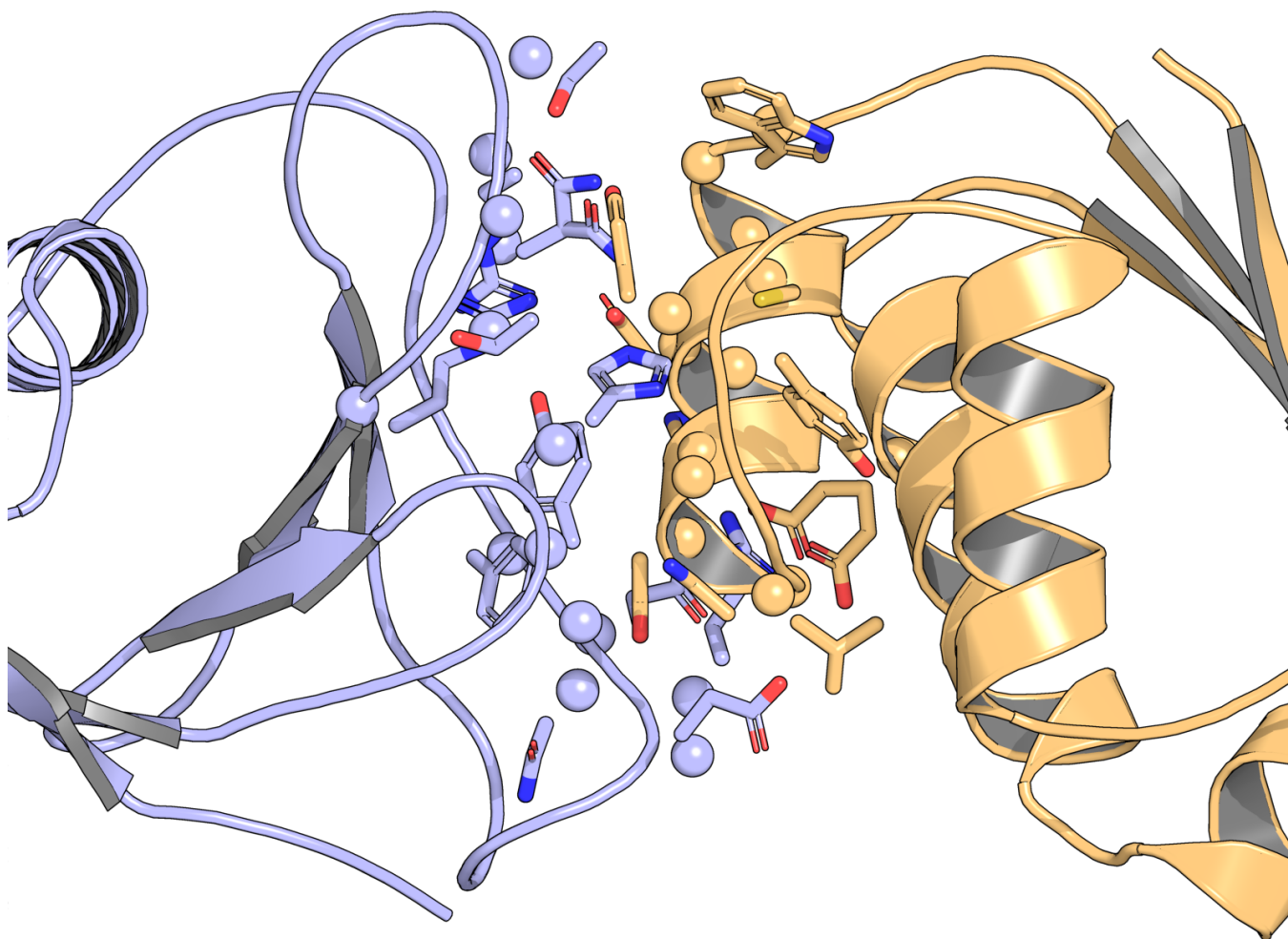
mutation site structural analysis



surface molecular surface per-chain

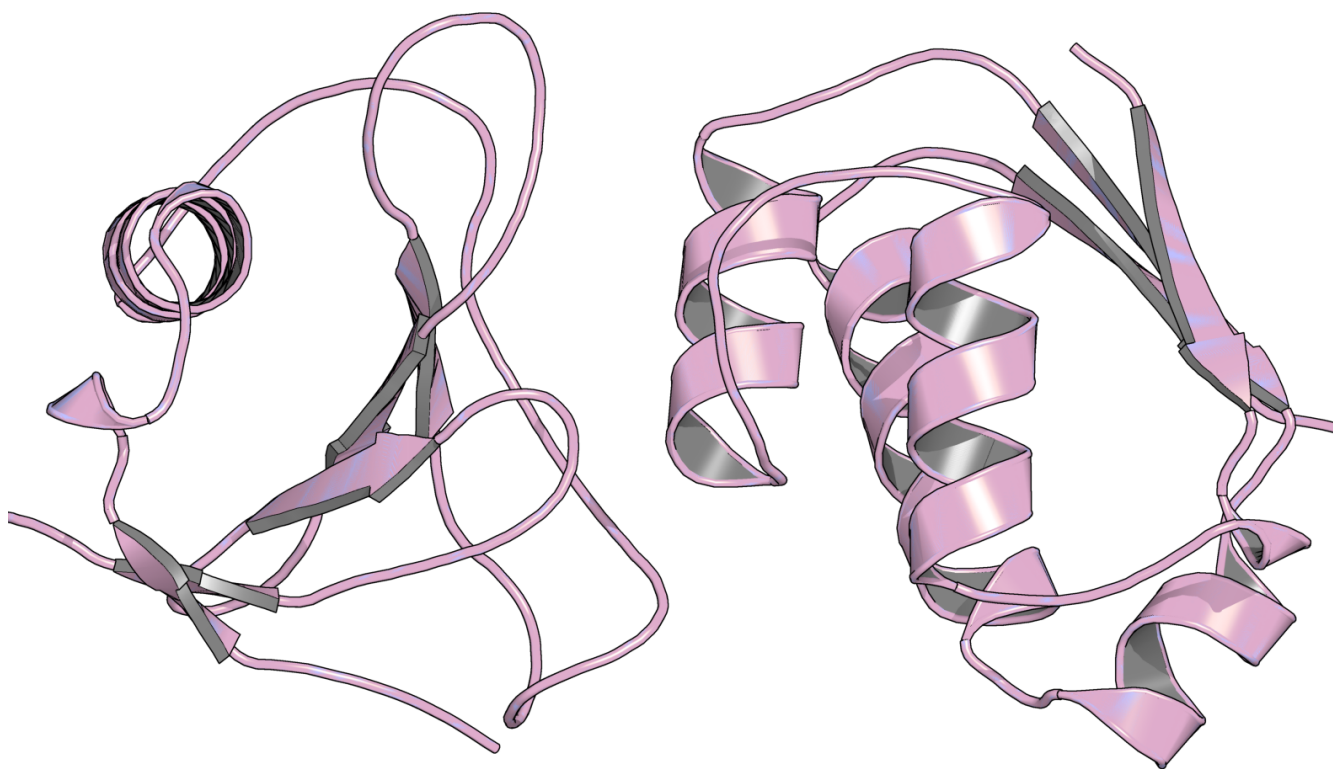


ensemble multi-model overlay



ppi protein-protein interface

Standard-class results



rmsd dual-structure overlay

Citation

Chen, Y., Zhang, H., Wang, W., Shen, Y., Ping, Z. (2024). Rapid generation of high-quality structure figures for publication with PyMOL-PUB. *Bioinformatics*, 40(3), btae139. <https://doi.org/10.1093/bioinformatics/btae139>